

Powering Down the Orion NanoFab Before Expected Power Outage

Occasionally, facilities require all equipment to be powered down for electrical maintenance. This procedure will step through the recommended process for safely powering down the NanoFab and then the steps needed to restore it to full operation.

In this procedure, we assume that powering down the NanoFab (including the pumps) will compromise the ultra-high vacuum level required for GFIS operation. Therefore, in order to prevent the formation of ice on the GFIS source, it is important that the GFIS gun and column are brought up to room temperature *before* compromising the GFIS vacuum level. This process requires several hours (depending on level of nitrogen in the dewar). The dewar warming process must be started the day *before* a scheduled power outage. For example, if power outage is expected on a Friday, begin the warming process on Thursday evening *before* the scheduled dewar autofill for that evening.

Please allow about one hour to perform the shut-down steps before the time of the expected power outage.

Sample Stage

- ✓ 1. Save a stage label appropriate for SFIM imaging and source maintenance (blank metal).
2. Unload the sample.

GFIS Column (Zen)

3. From Zen, place the GFIS column into Standby. This should:
 - ✓ a. Shut off the GFIS gas flow – helium or neon
 - ✓ b. Close the GFIS column isolation valve
 - ✓ c. Set the GFIS column to the Pumped state (turn on the GFIS ion pump)If any of these actions do not take place, please perform them manually.

Dewar (Note: this step must be performed one day *before* the scheduled power outage)

4. If the NanoFab software revision is Brisbane or earlier, the GFIS high voltages must be ramped down manually. This can be done in the Zen user interface.
5. From Service Machine (Tool's pull down menu from Athena System Manager), select GFIS Column Maintenance.
6. Select Warm to Room Temperature and click Execute Now. This should:
 - a. Ramp down all GFIS high voltage
 - b. Turn on the dewar heater (and other heaters) to begin warming the column and gun

Note: The GFIS high voltages will NOT disable. This can be done from Service Machine, Power Distribution, Power Supplies tab; otherwise, the high voltages will disable by interlock when the gun temperature rises.

Chamber

- ✓ 7. From Zen Operation Center, power down the ET and Flood Gun.
- ✓ 8. Disable the Sample Bias. (Note: Beginning with Damascus software, the sample bias is part of the ET detector and will power down with the detector.)

✓ NPVE

9. Exit the NPVE user interface.
10. Power down the NPVE controller/scan generator (white electronics box).
11. Shut down the NPVE PC.

X Capella Column

12. From Zen, power down the GaFIB (Capella column). This should:
 - a. Close the column isolation valve
 - b. Ramp down and disable all high voltageIf either of these actions do not take place, please perform them manually.
13. From the Vacuum screen, power off the lower (column) ion pump by right clicking the green square icon and selecting Turn Off
14. From Vacuum screen, right click on the GaFIB column state and select Connect. This should:
 - a. Open the bypass valve to the chamber
 - b. Shut off the upper (gun) ion pump

X GIS

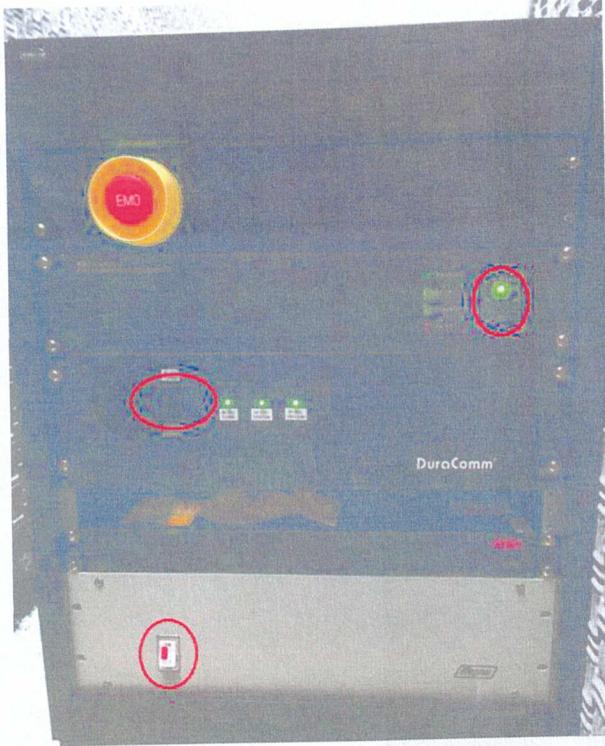
15. Retract the GIS nozzle.
16. Close the Oxford/Omni GIS user interface.
17. Power down the GIS controller from the power switch at the back of the controller.

Turbo Pumps

- ✓ 18. In the vacuum screen on the Dewar tab, right click and close the dewar jacket valve.
- ✓ 19. On the vacuum screen, check the Diag box at the lower right corner.
- ✓ 20. In the Airlock and Chamber tab of the vacuum screen, right click on the green square in chamber turbo pump and select Turn Off.
- ✓ 21. On the GFIS Column tab of the vacuum screen, right click on the green square in gun turbo pump and select Turn Off.
- ✓ 22. Allow the turbo pumps to shut down completely before doing the next step (the text will go from "braking" to "Off").
- ✓ 23. In the Airlock and chamber tab, close the chamber foreline valve.
- ✓ 24. In the Airlock and chamber tab, close the chamber safety valve and select Turn Off by right clicking the chamber rough pump icon.
- ✓ 25. In the GFIS Column tab, close the gun safety valve and select Turn Off by right clicking the gun rough pump icon.
- ✓ 26. In the Dewar tab, close the dewar rough valve.

Electronics Rack, PC and Chase Rack

27. Open the Power Distribution page of Service machine and uncheck all boxes on the right side of the screen.
28. Exit the Zen user interface.
29. Press the Stop button on the Athena System Manager and wait for the message "Server is shut down" to appear in the window.
30. Shut down the NanoFab PC.
31. At the chase rack power off the following four switches (see image below):
 - a. The XEI Scientific Evactron Decontaminator controller
 - b. Two switches on the Duracomm power supply
 - c. One switch on the Acopian power supply



32. Press the EMO button on the Chase Rack and switch off both circuit breakers on the back of the UPS



33. Shut off the dewar pump (Sogevac) by pushing the red Stop button.



The system is now ready for the facilities shut down.

- ① CA2 cylinder off
- ② N₂ " "
- ③ LN₂ dewar off.

Powering Up the Orion NanoFab After Restoring Electrical Power

Returning the NanoFab to operational state is *not* necessarily a matter of reversing the above steps.

Facilities

Verify the facilities (CDA & N2) are back on and at the desired pressures.

Electronics Rack, PC and Chase Rack

1. Unlock the EMO button on the chase rack.
2. Switch the Sogevac dewar pump on by pressing the Start button.
3. Set both circuit breakers on the rear of the UPS to the ON position and allow the UPS to perform its self checks. After about one minute, the display on the UPS should read "ONLINE MODE".
4. Turn on the power supplies in the chase rack; these are the same four switches as in step 31 of the shut-down procedure above. Note: the light on the Acopian power switch will not illuminate yet.
5. Turn on the NanoFab PC and allow it to boot up.
6. Log in, open and start the Athena System Manager; wait for the message "The server has started".
7. Verify the Power Distribution page of Service Machine has restarted (all the boxes that needed to be unchecked earlier, should now appear checked. Normally, only the Dewar Bake should remain unchecked). If not, check each box on the right side except the Dewar Bake checkbox.
8. Wait five minutes for all the circuit boards to boot up.
9. Open the Hardware Communications Status page of Service Machine and verify all modules are talking (no yellow or red highlights).
10. Stop and restart the Athena System Manager if necessary to re-establish communications to all the modules.
11. Open the Firmware Update page of Service Machine and click on the Query button. All should respond with a green check mark.
12. From the Athena System Manager Tools pull down menu, select vacuum screen.
13. Start the Zen user interface.

Dewar

14. In the vacuum screen Dewar tab, verify that the dewar state is Idle (this state will not allow the dewar to fill).

Capella Column

15. Right click the state of the GaFIB column on the vacuum screen and select Connect.

Vacuum

16. In the vacuum screen, right click the GFIS Column vacuum state and select Pump No Ion Pump. The column roughing pump should turn on and the associated valves will cycle as needed. Eventually the gun turbo pump will turn on and spin up but the GFIS ion pump will not attempt to turn on.
17. In the vacuum screen, right click on the Chamber state and select Rough. The chamber roughing pump should turn on and the associated valves will cycle as needed.
18. When the chamber has achieved the Rough Vac state, open the dewar jacket valve in the Dewar tab.
19. Select Pump in the chamber state, eventually the chamber turbo will spin up and change to Ready.

Capella Column

20. From the Zen user interface, home the apertures for the GaFIB column.
21. When the chamber has reached the -7T range, turn on the lower (column) ion pump. Monitor the lower (column) ion pump pressure to make sure it is decreasing.

It is recommended that the GaFIB gun be pumped by the chamber turbo through the bypass valve for twelve hours before turning on the gun ion pump and closing the bypass valve. After this twelve hours of pumping, continue with step 22 below.

22. From the GaFIB state on the vacuum screen, select isolate. This will turn on the upper (gun) ion pump and close the bypass valve to the chamber. Monitor the upper (gun) ion pump pressure to make sure it is decreasing.

GIS

23. Switch on the Oxford/OmniGISII controller.
24. Start the Oxford/OmniGISII software.
25. Home the GIS nozzle and allow it to go to its Retracted position.

Chamber

26. Verify that the chamber is in the Pumped state.
27. From the Zen user interface, home the stage.
28. From the Zen Operation Center, power up the ET detector.

GFIS Column

The GFIS gun and column will require a bake. The standard recommended bake time is 100 hours.

29. Verify that the GFIS column state is Pumped No Ion Pump.
30. From the Zen user interface, home the apertures for the GFIS column.
31. From Service Machine, GFIS Column Maintenance, select "Warm, Bake and Get Cold" and set the number of hours to bake. Make sure that the liquid nitrogen supply is connected and that the tank valve is open so that the dewar will fill after the bake has completed.
32. Click Execute Now.

After the bake is completed, the recommended recovery for the GFIS source is a Light Anneal followed by a Maintain BIV source build.

33. Verify that the GFIS column state is Pumped (with the GFIS ion pump on).
34. Verify that the gun pressure has reached the -10T range -- this is the threshold at which the software will allow helium flow into the gun so that trimer formation can be performed.
35. In Service Machine, select Power Distribution and Power Supplies tab.
36. Enable each of the four GFIS high voltage elements by checking the boxes.
37. Load the sample holder with an appropriate sample for SFIM imaging (blank metal) and drive the stage to the saved label for SFIM imaging.
38. Perform a Light Anneal from the Zen GFIS Column Maintenance menu.
39. After the light anneal has completed, perform a Maintain BIV build from the Zen GFIS Column Maintenance menu.
40. Continue with trimer formation and column alignments in the normal manner to fully recover the GFIS column.

NPVE

41. Power up the NPVE Controller box and wait one minute.
42. Power up the NPVE computer and allow it to boot up.

Connect

HH cables

43. On the *NanoFab PC (NOT the NPVE PC)*, start the ZeissFibicsServer software. This may appear as an icon pinned to the Start menu but can also be started by double clicking the FibicsZeissServer.exe file in the following folder:

C:\Program Files (x86)\Fibics\Server

44. Log in and start the NPVE user interface.

